

4DVarNet-FM Synthesis Report

4DVarNet-FM: Experiment Synthesis Report (live-updated)

Generated: 2026-07-01 13:03:01 | T_max = 3.0s | dt = 0.01 | 15 obs/window

Part 1 – 4DVarNet-FM Variants:

Config	CS1 X	CS1 Y	CS1 Z	CS1 μ	CS2 X	CS2 Y	CS2 Z	CS2 μ	Deg	Time
B1_small_unet	0.081	0.098	0.113	0.097	0.100	0.109	0.122	0.110	1.13	x 47m
C1_longer_train	0.172	0.165	0.190	0.175	0.178	0.166	0.192	0.179	1.02	x 105m
C4_stage1_only	5.239	9.310	23.128	12.559	5.340	9.312	22.558	12.404	0.99	x 5m
D1_cs2_only	7.754	6.510	7.441	7.235	7.247	6.606	7.598	7.150	0.99	x 33m
E1_direct_unet_default	0.155	0.140	0.138	0.144	0.173	0.158	0.138	0.156	1.08	x 8m
E2_direct_unet_small	0.047	0.072	0.125	0.081	0.073	0.091	0.142	0.102	1.25	x 7m
E3_direct_unet_rand	0.133	0.135	0.081	0.116	0.131	0.130	0.081	0.114	0.98	x 7m
F1_vanilla_cfm_default	0.109	0.153	0.159	0.141	0.105	0.166	0.175	0.149	1.06	x 16m
F2_vanilla_cfm_small	0.066	0.133	0.158	0.119	0.075	0.150	0.152	0.126	1.06	x 12m
F3_vanilla_cfm_rand	0.049	0.071	0.087	0.069	0.051	0.069	0.090	0.070	1.01	x 13m

Part 2 – DA Baselines (same test data):

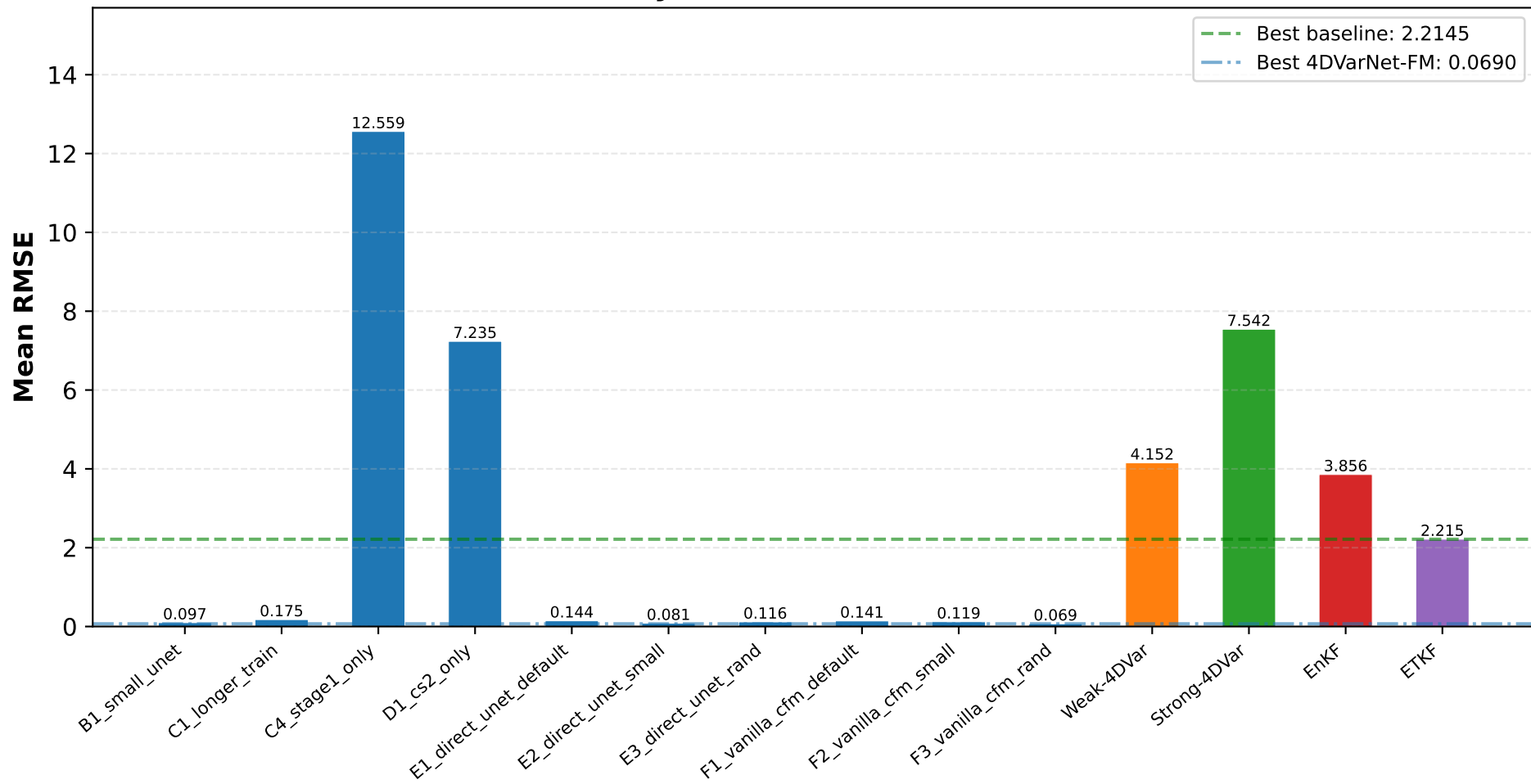
Method	CS1 μ	CS2 μ	Deg	
Weak-4DVar	4.6450	4.1519	0.89	x
Strong-4DVar	6.1428	7.5424	1.23	x
EnKF	1.2381	3.8564	3.11	x
ETKF	0.6888	2.2145	3.22	x

Best 4DVarNet-FM: F3_vanilla_cfm_rand (CS2 μ =0.0700)

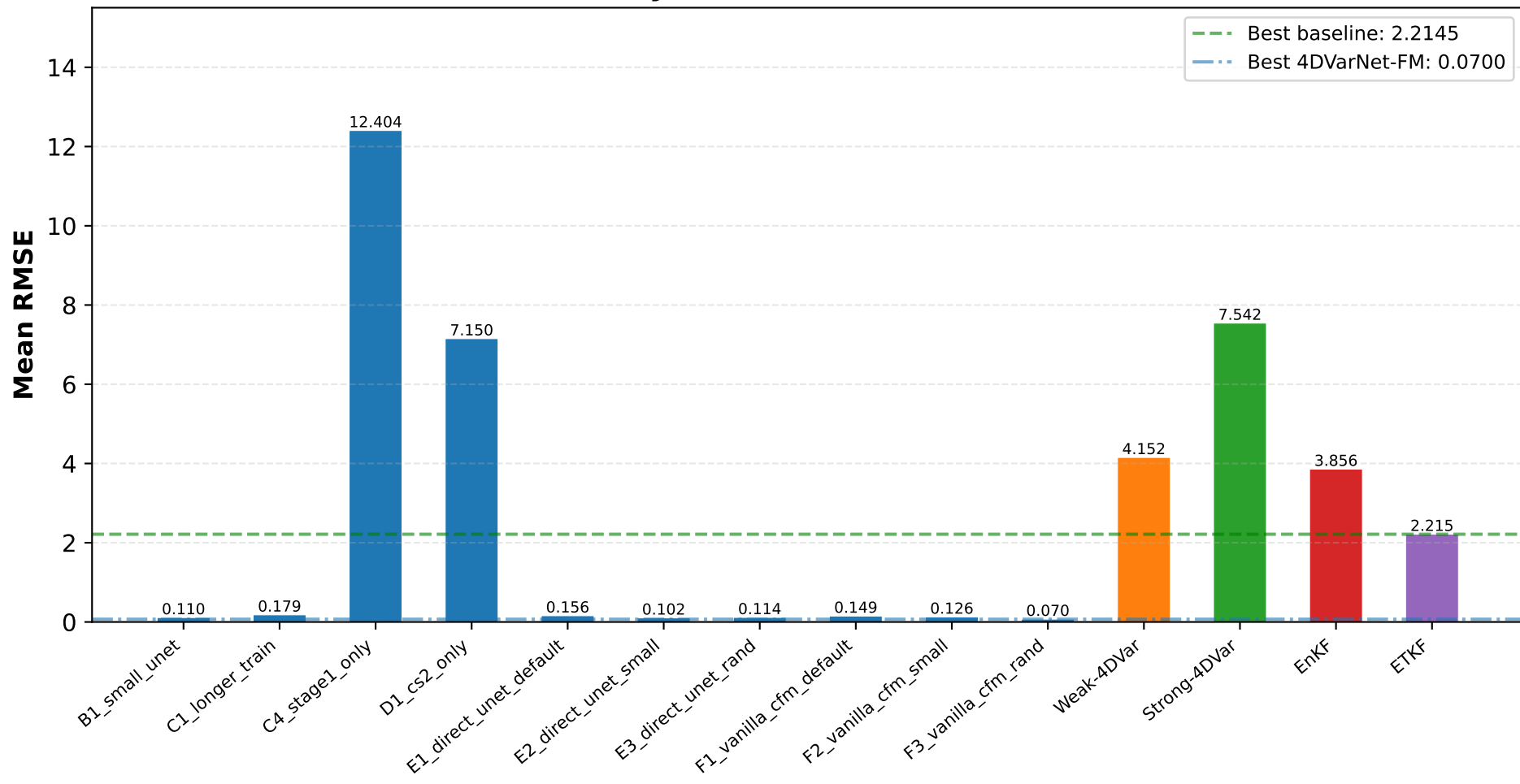
Best baseline CS1: ETKF μ =0.6888

Best baseline CS2: ETKF μ =2.2145

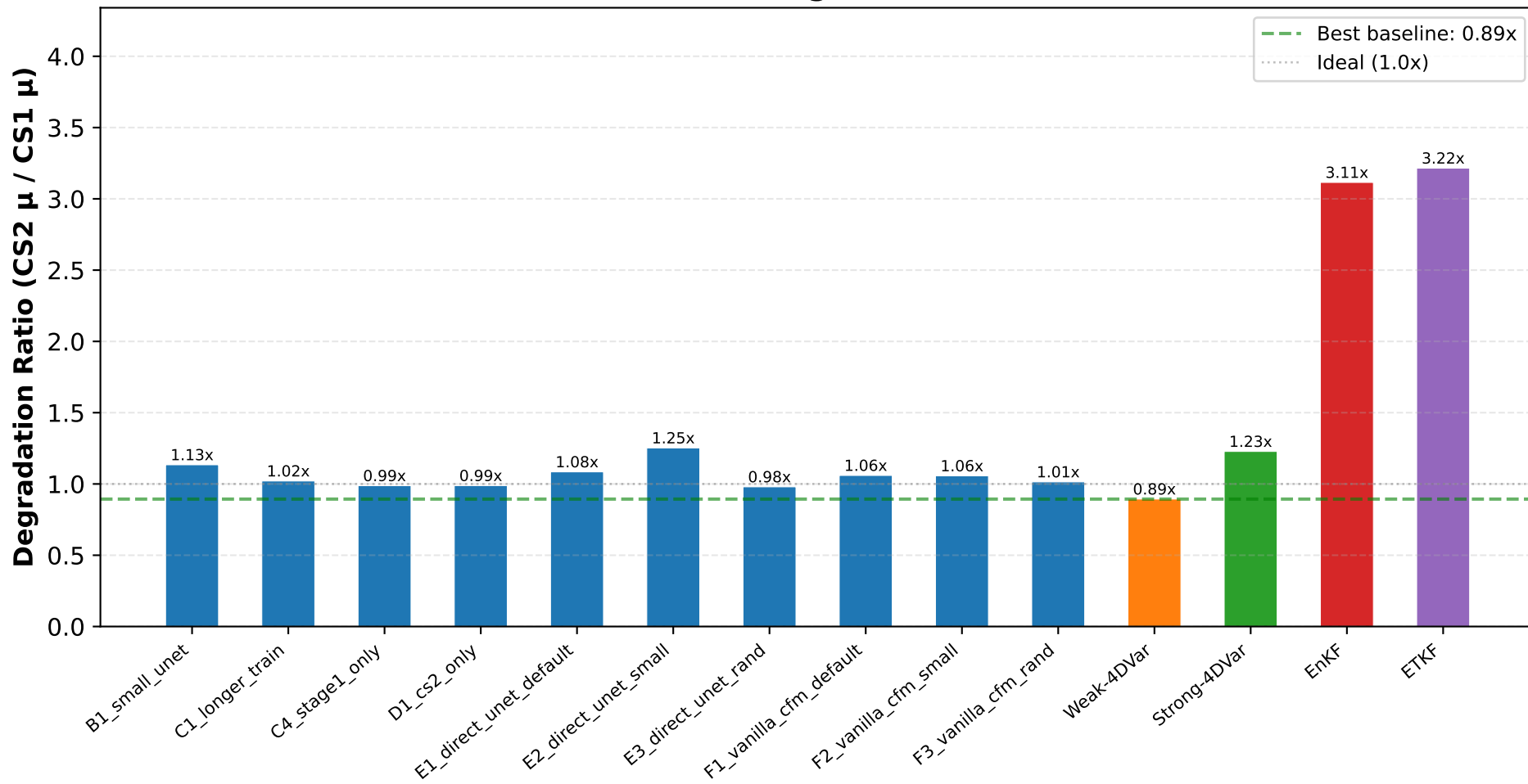
Case Study 1 — RMSE (lower is better)



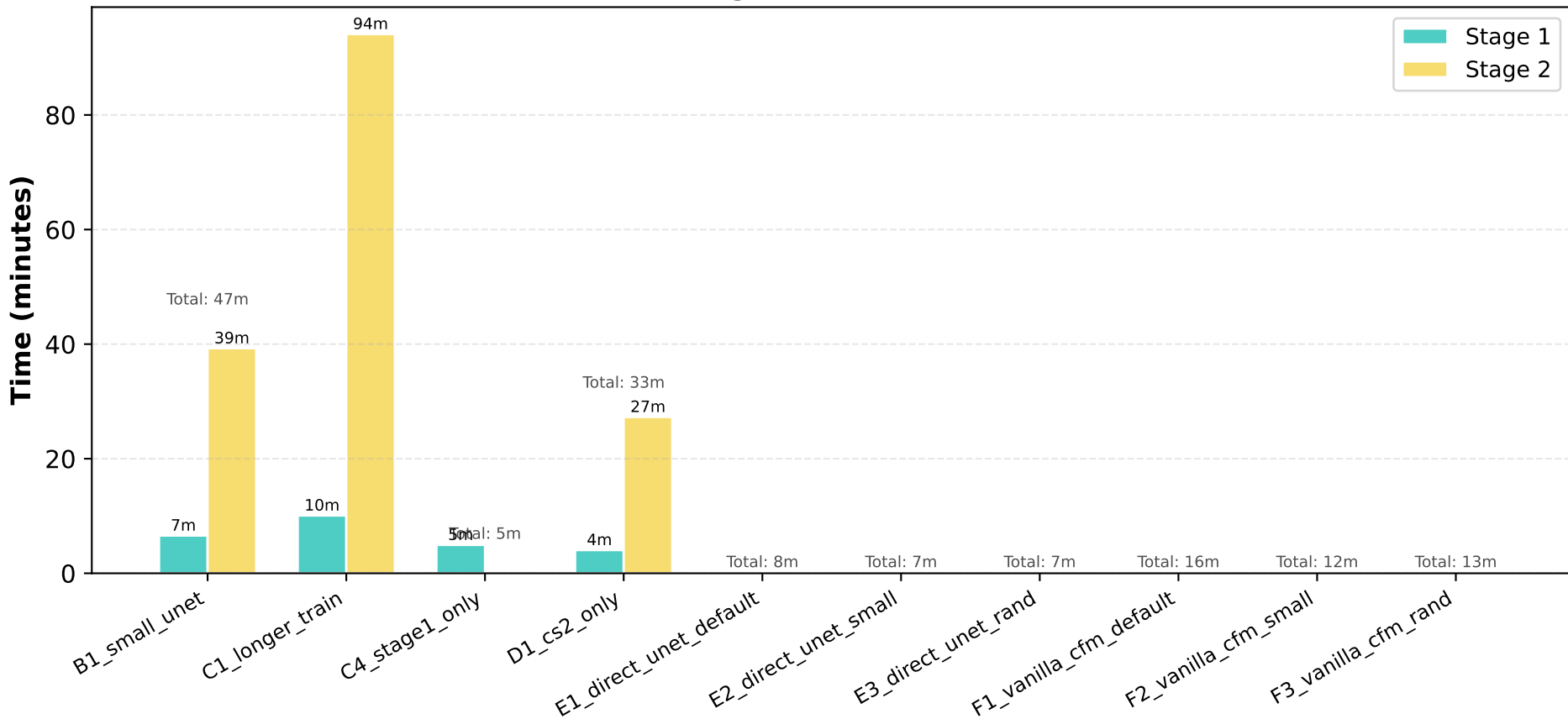
Case Study 2 — RMSE (lower is better)



Robustness — Lower Degradation = More Robust



Training Time Breakdown



4DVarNet-FM Synthesis Report — Conclusion

Conclusion & Recommendations (live-updated)

Best 4DVarNet-FM variant: F3_vanilla_cfm_rand
CS1 mean RMSE: 0.0690 (best baseline ETKF: 0.6888) ✓ BEATS baseline
CS2 mean RMSE: 0.0700 (best baseline ETKF: 2.2145) ✓ BEATS baseline
Degradation: 1.01x

Comparison across all variants:

B1_small_unet	CS1 $\mu=0.0973$	CS2 $\mu=0.1104$	Deg=1.13x	★ BEATS ALL BASELINES
C1_longer_train	CS1 $\mu=0.1753$	CS2 $\mu=0.1788$	Deg=1.02x	★ BEATS ALL BASELINES
C4_stage1_only	CS1 $\mu=12.5590$	CS2 $\mu=12.4035$	Deg=0.99x	Below baselines
D1_cs2_only	CS1 $\mu=7.2349$	CS2 $\mu=7.1504$	Deg=0.99x	Below baselines
E1_direct_unet_default	CS1 $\mu=0.1441$	CS2 $\mu=0.1563$	Deg=1.08x	★ BEATS ALL BASELINES
E2_direct_unet_small	CS1 $\mu=0.0815$	CS2 $\mu=0.1020$	Deg=1.25x	★ BEATS ALL BASELINES
E3_direct_unet_rand	CS1 $\mu=0.1163$	CS2 $\mu=0.1139$	Deg=0.98x	★ BEATS ALL BASELINES
F1_vanilla_cfm_default	CS1 $\mu=0.1407$	CS2 $\mu=0.1491$	Deg=1.06x	★ BEATS ALL BASELINES
F2_vanilla_cfm_small	CS1 $\mu=0.1190$	CS2 $\mu=0.1257$	Deg=1.06x	★ BEATS ALL BASELINES
F3_vanilla_cfm_rand	CS1 $\mu=0.0690$	CS2 $\mu=0.0700$	Deg=1.01x	★ BEATS ALL BASELINES

Key Insights:

Recommendation:
Use F3_vanilla_cfm_rand for production (lowest CS2 RMSE).